

INTERNATIONAL CYBERSECURITY AND DIGITAL FORENSICS ACADEMY (ICDFA)

Case Study 1 - Individual Foensic Investigation Investigating Harassment Email Traffic With Wireshark

Lab Title:	Investigating Harassment Email Traffic With Wireshark
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Course Code:	SBT-DF204
Course Name:	Computer Forensics Case Studies

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Case scenario

Lily Tuckrige, a Chemistry Department teacher, reports repeated harassing emails sent to her personal Yahoo Mail account. Full headers from an earlier message point to an IP address in a shared student residence where an open wireless router is in use.

On Monday 21 July, a later message arrived through the web service willselfdestruct.com. The service briefly displayed the message and then reported that it had been destroyed. The university captured traffic at the residence network boundary and asks you to determine whether the available evidence supports attribution to a student in Chemistry 109.

2. Required Tools and Environment

Item	Recommended Tool	Purpose
Operating system	Kali Linux or any Wireshark-capable OS	Analyze the capture.
Packet tools	Wireshark and tshark	Filter, inspect, and follow streams.
Hashing	sha256sum	Preserve evidence integrity.
Evidence	nitroba.pcap	Publicly published training capture (digitalcorpora.org).
Reference material	Scenario page, class roster	Cross-check identity/attribution questions.

3. Case Folder Setup and Acquire Evidence

```
(kali@kali)-[~]
└─$ mkdir -p ~/SBT-DF204-Case1/{evidence,working,exported,reports,screenshots}
cd ~/SBT-DF204-Case1
pwd
/home/kali/SBT-DF204-Case1
find . -maxdepth 1 -type d -print
./exported
./evidence
./working
./screenshots
./reports
```

```
(kali@kali)-[~/SBT-DF204-Case1]
└─$ wget -O evidence/nitroba.pcap \
    'https://digitalcorpora.s3.amazonaws.com/corpora/scenarios/2008-nitroba/nitroba.pcap'
--2026-09-30 19:00:56-- https://digitalcorpora.s3.amazonaws.com/corpora/scenarios/2008-nitroba/nitroba.pcap
Resolving digitalcorpora.s3.amazonaws.com (digitalcorpora.s3.amazonaws.com)... 52.92.155.105, 52.92.235.33, 3.5.84.60, ...
Connecting to digitalcorpora.s3.amazonaws.com (digitalcorpora.s3.amazonaws.com)|52.92.155.105|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 56180821 (54M) [binary/octet-stream]
Saving to: 'evidence/nitroba.pcap'

evidence/nitroba.pcap      100%[=====] 53.58M  872KB/s  in 3m 23s

2026-09-30 19:04:21 (270 KB/s) - 'evidence/nitroba.pcap' saved [56180821/56180821]
```

```
(kali@kali)-[~/SBT-DF204-Case1]
└─$ sha256sum evidence/nitroba.pcap | tee reports/nitroba_sha256.txt
2b77a9eaefc1d6af163d1ba793c96dbccac04e6befdf1a0b01f8c67553ec2fb evidence/nitroba.pcap

(kali@kali)-[~/SBT-DF204-Case1]
└─$ cp --preserve=timestamps evidence/nitroba.pcap working/nitroba_working.pcap
sha256sum working/nitroba_working.pcap | tee -a reports/nitroba_sha256.txt
ls -la evidence/nitroba.pcap
2b77a9eaefc1d6af163d1ba793c96dbccac04e6befdf1a0b01f8c67553ec2fb working/nitroba_working.pcap
-rw-rw-r-- 1 kali kali 56180821 Nov 21 2020 evidence/nitroba.pcap
```

```

(kali@kali)-[~/SBT-DF204-Case1]
$ capinfos evidence/nitroba.pcap | tee reports/capinfos.txt
File name:          evidence/nitroba.pcap
File type:          Wireshark/tcpdump/... - pcap
File encapsulation: Ethernet
File timestamp precision: microseconds (6)
Packet size limit:  file hdr: 4096 bytes
Number of packets:  94 k
File size:          56 MB
Data size:          54 MB
Capture duration:   15759.950751 seconds
Earliest packet time: 2008-07-21 21:51:07.095278
Latest packet time:  2008-07-22 02:13:47.046029
Data byte rate:     3,468 bytes/s
Data bit rate:      27 kbps
Average packet size: 579.07 bytes
Average packet rate: 5 packets/s
SHA256:             2b77a9eaefc1d6af163d1ba793c96dbccacb04e6befdf1a0b01f8c67553ec2fb
SHA1:               65656392412add15f93f8585197a8998aaeb50a1
Strict time order:  True
Number of interfaces in file: 1
Interface #0 info:
    Encapsulation = Ethernet (1 - ether)
    Capture length = 4096
    Time precision = microseconds (6)
    Time ticks per second = 1000000
    Number of stat entries = 0
    Number of packets = 94410

```

wget downloads the official, publicly hosted case file directly from digitalcorpora.org's S3 bucket into evidence/(folder) as untouched original. sha256sum fingerprints it immediately. The calculated SHA-256 value here becomes the reference hash which another examiner can use to confirm, if they're looking at the same file. cp makes my working copy, capinfos gives me the file size, packet count, and duration.

4. Evidence Acquisition and Integrity - Report Section

Field	Entry
File name	nitroba.pcap
Source URL	digitalcorpora.s3.amazonaws.com/corpora/scenarios/2008-nitroba/nitroba.pcap
File size	56,180,821 bytes (56 MB / 53.58 MiB)
Download date	2026-09-30
Calculated SHA-256 (original)	2b77a9eaefc1d6af163d1ba793c96dbccacb04e6befdf1a0b01f8c67553ec2fb
Working-copy SHA-256	2b77a9eaefc1d6af163d1ba793c96dbccacb04e6befdf1a0b01f8c67553ec2fb
Published checksum comparison	Not published on source page at time of writing -9 calculated value recorded above for examiner reference
Analysis workstation	Kali

5. Part A -Inventory the Capture and Locate the Web Service

```
(kali@kali)-[~/SBT-DF204-Case1]
$ capinfos working/nitroba_working.pcap | tee reports/capinfos_working.txt
tshark -r working/nitroba_working.pcap -q -z io,phs | tee reports/protocol_hierarchy.txt
tshark -r working/nitroba_working.pcap -Y "http.request.method == \"GET\" && http.host == \"www.willselfdestruct.com\"" \
-T fields -e frame.number -e frame.time -e ip.src -e ip.dst -e http.host -e http.request.uri \
| tee reports/willselfdestruct_requests.tsv
File name: working/nitroba_working.pcap
File type: Wireshark/tcpdump/... - pcap
File encapsulation: Ethernet
File timestamp precision: microseconds (6)
Packet size limit: file hdr: 4096 bytes
Number of packets: 94 k
File size: 56 MB
Data size: 54 MB
Capture duration: 15759.950751 seconds
Earliest packet time: 2008-07-21 21:51:07.095278
Latest packet time: 2008-07-22 02:13:47.046029
Data byte rate: 3,468 bytes/s
Data bit rate: 27 kbps
Average packet size: 579.07 bytes
Average packet rate: 5 packets/s
SHA256: 2b77a9eafc1d6af163d1ba793c96dbccac04e6ebdf1a0b01f8c67553ec2fb
SHA1: 65656392412add15f93f8585197a8998aaeb50a1
Strict time order: True
Number of interfaces in file: 1
Interface #0 info:
    Encapsulation = Ethernet (1 - ether)
    Capture length = 4096
    Time precision = microseconds (6)
    Time ticks per second = 1000000
```

Number of stat entries = 0
Number of packets = 94410

Protocol Hierarchy Statistics

Filter:

frame	frames:94410 bytes:54670237
eth	frames:94410 bytes:54670237
ip	frames:90379 bytes:54413507
tcp	frames:83114 bytes:52515171
http	frames:10103 bytes:7007064
image-gif	frames:1136 bytes:737880
image-jfif	frames:1180 bytes:982489
data-text-lines	frames:744 bytes:529526
xml	frames:581 bytes:554874
png	frames:136 bytes:112564
http	frames:4 bytes:1484
urlencoded-form	frames:28 bytes:14515
media	frames:467 bytes:248381
data	frames:7 bytes:7097
amf	frames:1 bytes:511
ocsp	frames:2 bytes:1108
tls	frames:1561 bytes:931191
tls	frames:216 bytes:245641
vssmonitoring	frames:1 bytes:64
_ws.malformed	frames:1 bytes:64
prp	frames:2 bytes:134
xmpp	frames:18 bytes:3131
data	frames:50 bytes:12304


```

      ntp          frames:184 bytes:17296
      ssdp         frames:2225 bytes:818614
      dcerpc       frames:23 bytes:13487
      messenger    frames:22 bytes:13065
      bt-dht       frames:38 bytes:5495
      rtp          frames:1060 bytes:230772
      rtcp         frames:12 bytes:1372
      isakmp       frames:2 bytes:1411
      udpencap     frames:448 bytes:148734
      isakmp       frames:40 bytes:7952
      esp          frames:395 bytes:139950
      rip          frames:2 bytes:140
      data         frames:4 bytes:472
      icmp         frames:20 bytes:21396
      igmp         frames:45 bytes:2592
      arp          frames:4026 bytes:256410
      llc          frames:5 bytes:320
      data         frames:5 bytes:320

82936 2008-07-22T02:03:43.825871000-0400 192.168.15.4 69.25.94.22 www.willselfdestruct.com /secure/submit
82985 2008-07-22T02:03:44.125949000-0400 192.168.15.4 69.25.94.22 www.willselfdestruct.com /images/spacer.gif
83025 2008-07-22T02:03:44.334254000-0400 192.168.15.4 69.25.94.22 www.willselfdestruct.com /images/sm-logo.gif
83037 2008-07-22T02:03:44.374583000-0400 192.168.15.4 69.25.94.22 www.willselfdestruct.com /images/warning-home.gif
83072 2008-07-22T02:03:44.530352000-0400 192.168.15.4 69.25.94.22 www.willselfdestruct.com /images/body-bk.gif
83087 2008-07-22T02:03:44.599079000-0400 192.168.15.4 69.25.94.22 www.willselfdestruct.com /images/btn-send.gif
83162 2008-07-22T02:03:44.852265000-0400 192.168.15.4 69.25.94.22 www.willselfdestruct.com /images/bridge_small.gif
83614 2008-07-22T02:04:24.564165000-0400 192.168.15.4 69.25.94.22 www.willselfdestruct.com /secure/success
83654 2008-07-22T02:04:24.902157000-0400 192.168.15.4 69.25.94.22 www.willselfdestruct.com /images/bk-message.gif

(kali@kali) ~/SBT-DF204-Case1
$ tshark -r working/nitroba_working.pcap -Y "ip.addr==69.25.94.22 && http.request" -T fields \
-e frame.number -e frame.time -e http.request.method -e http.request.uri -e http.request.full_uri \
tee reports/willselfdestruct_all_requests.tsv
82936 2008-07-22T02:03:43.825871000-0400 GET /secure/submit http://www.willselfdestruct.com/secure/submit
82985 2008-07-22T02:03:44.125949000-0400 GET /images/spacer.gif http://www.willselfdestruct.com/images/spacer.gif
83025 2008-07-22T02:03:44.334254000-0400 GET /images/sm-logo.gif http://www.willselfdestruct.com/images/sm-logo.gif
83037 2008-07-22T02:03:44.374583000-0400 GET /images/warning-home.gif http://www.willselfdestruct.com/images/warning-hom
e.gif
83072 2008-07-22T02:03:44.530352000-0400 GET /images/body-bk.gif http://www.willselfdestruct.com/images/body-bk.gif
83087 2008-07-22T02:03:44.599079000-0400 GET /images/btn-send.gif http://www.willselfdestruct.com/images/btn-send.gif
83162 2008-07-22T02:03:44.852265000-0400 GET /images/bridge_small.gif http://www.willselfdestruct.com/images/bridge_smal
l.gif
83601 2008-07-22T02:04:24.311700000-0400 POST /secure/submit http://www.willselfdestruct.com/secure/submit
83614 2008-07-22T02:04:24.564165000-0400 GET /secure/success http://www.willselfdestruct.com/secure/success
83654 2008-07-22T02:04:24.902157000-0400 GET /images/bk-message.gif http://www.willselfdestruct.com/images/bk-message.gif

```

The protocol hierarchy gives the overview of everything in the capture (HTTP, DNS, SMTP, etc.), to observe before diving into specific filters. The second filter is the brief's suggested starting point, which isolates every GET request made specifically to www.willselfdestruct.com, the self-destructing message service named in the scenario. This confirms whether/when the client actually visited the service's page, which serves as my first concrete link between the capture and the case narrative.

Item	My Findings
Total packets / capture duration	94,410 packets total (confirmed in protocol hierarchy); duration ~4h 22m (per earlier capinfos: 21:51:07 to 02:13:47)
Client IP contacting the service	192.168.15.4
Service IP (willselfdestruct.com)	69.25.94.22
Packet number of first GET request	82936
Timestamp of first GET request	2008-07-22T02:03:43.825871000-0400
Packet number of the relevant POST	83601
Timestamp of the POST	2008-07-22T02:04:24.311700000-0400
URI	/secure/submit
Full URL	http://www.willselfdestruct.com/secure/submit

5. Part B - Search Page Content and Confirm the Service

```
(kali@kali)-[~/SBT-DF204-Case1]
$ tshark -r working/nitroba_working.pcap -Y "data-text-lines contains \"secure anonymous E-mail\"" \
-T fields -e frame.number -e frame.time -e ip.src -e ip.dst \
| tee reports/service_content_match.tsv
82998 2008-07-22T02:03:44.210946000-0400 69.25.94.22 192.168.15.4
83641 2008-07-22T02:04:24.851431000-0400 69.25.94.22 192.168.15.4
```

This searches reassembled HTTP response text for the phrase associated with the willselfdestruct.com service's own page copy. A match here corroborates, from the actual page content returned to the client, that this really is the self-destructing-message service described in the case scenario, not just a coincidentally-named host.

Item	Findings
Packet number(s)	82998 and 83641
Timestamp	2008-07-22T02:03:44.210946000-0400 (first); 2008-07-22T02:04:24.851431000-0400 (second)
Source → Destination	69.25.94.22 (willselfdestruct.com) → 192.168.15.4 (client)
What this confirms	The server's own page content contains the phrase "secure anonymous E-mail" directly verifying this is the self-destructing-message service described in the case scenario, confirmed from the actual page text returned to the client, not just the hostname.

6. Part C - Find the HTTP POST (the Message Submission)

```
(kali@kali)-[~/SBT-DF204-Case1]
$ tshark -r working/nitroba_working.pcap -Y "http.request.method == \"POST\"" \
-T fields -e frame.number -e frame.time -e ip.src -e tcp.srcport -e ip.dst -e tcp.dstport \
-e http.host -e http.request.uri \
| tee reports/http_posts.tsv
10217 2008-07-21T21:56:23.170447000-0400 192.168.1.64 46912 64.12.187.25 80 sync.aol.com /m57jean
10241 2008-07-21T21:56:23.335872000-0400 192.168.1.64 46912 64.12.187.25 80 sync.aol.com /m57jean
10248 2008-07-21T21:56:23.580097000-0400 192.168.1.64 46912 64.12.187.25 80 sync.aol.com /m57jean
10869 2008-07-21T21:59:54.595252000-0400 192.168.1.64 43821 64.4.21.61 80 www.update.microsoft.com /v6/Update
17299 2008-07-21T23:44:46.722832000-0400 192.168.1.64 42634 74.125.19.19 80 mail.google.com /mail/?ui=26ik=4233eca8e46
view=au&rt=j
17305 2008-07-21T23:44:48.898068000-0400 192.168.1.64 42634 74.125.19.19 80 mail.google.com /mail/?ui=26ik=4233eca8e46
view=cv6th=11b48086a8a331ad6prf=16init=16nsc=16rt=j&search=inbox
17316 2008-07-21T23:44:49.893979000-0400 192.168.1.64 42634 74.125.19.19 80 mail.google.com /mail/?ui=26ik=4233eca8e46
view=cv6th=11b482d8bc5024226prf=16nsc=16rt=j&search=inbox
17373 2008-07-21T23:44:55.683957000-0400 192.168.1.64 42634 74.125.19.19 80 mail.google.com /mail/channel/bind?at=xn3j
3022594piz2yqigx80qxagfx76VER=66it=23596RID=539966CVER=66zx=kg7q7akmj9398t=1
17384 2008-07-21T23:44:57.766721000-0400 192.168.1.64 40694 74.125.19.19 80 mail.google.com /mail/channel/bind?at=xn3j
3022594piz2yqigx80qxagfx76VER=66it=20006SID=3FF4ADAE839696306RID=539976zx=9nresyky19hs&t=1
17422 2008-07-21T23:45:04.404751000-0400 192.168.1.64 42634 74.125.19.19 80 mail.google.com /mail/channel/bind?at=xn3j
3022594piz2yqigx80qxagfx76VER=66it=18736SID=3FF4ADAE839696306RID=539988zx=32uwmci9zj16t=1
17449 2008-07-21T23:45:06.841012000-0400 192.168.1.64 42634 74.125.19.19 80 mail.google.com /mail/channel/bind?at=xn3j
3022594piz2yqigx80qxagfx76VER=66it=96SID=3FF4ADAE839696306RID=539996zx=8jq2dwxflmv4&t=1
22531 2008-07-22T00:36:45.720384000-0400 192.168.15.4 32992 209.53.113.23 80 search.namequery.com:80 /
22534 2008-07-22T00:36:45.858943000-0400 192.168.15.4 32992 209.53.113.23 80 search.namequery.com:80 /
22537 2008-07-22T00:36:45.895435000-0400 192.168.15.4 32992 209.53.113.23 80 search.namequery.com:80 /
22540 2008-07-22T00:36:45.933309000-0400 192.168.15.4 32992 209.53.113.23 80 search.namequery.com:80 /
```

```
(kali@kali)-[~/SBT-DF204-Case1]
$ tshark -r working/nitroba_working.pcap -Y "frame.number==83601" -T fields -e tcp.stream
1707

(kali@kali)-[~/SBT-DF204-Case1]
$ tshark -r working/nitroba_working.pcap -q -z follow,tcp,ascii,1707 | tee reports/post_stream_content.txt
```

```

Follow: tcp,ascii
Filter: tcp.stream eq 1707
Node 0: 192.168.15.4:36044
Node 1: 69.25.94.22:80
649
POST /secure/submit HTTP/1.1
Accept: image/gif, image/x-bitmap, image/jpeg, image/pjpeg, application/x-shockwave-flash, */*
Referer: http://www.willselfdestruct.com/secure/submit
Accept-Language: en-us
Content-Type: application/x-www-form-urlencoded
Accept-Encoding: gzip, deflate
User-Agent: Mozilla/4.0 (compatible; MSIE 6.0; Windows NT 5.1; SV1)
Host: www.willselfdestruct.com
Content-Length: 188
Connection: Keep-Alive
Cache-Control: no-cache

to=lilytuckrige@yahoo.com&from=6subject=you+can%27t+find+us&message=and+you+can%27t+hide+from+us.%0D%0A%0D%0AStop+teaching.%0D%0A%0D%0AStart+running.+&type=0&ttl=30&submit.x=92&submit.y=26
289
HTTP/1.1 302 Moved Temporarily
Date: Tue, 22 Jul 2008 07:24:45 GMT
Server: Apache
Set-Cookie: JSESSIONID=C16BFAC8710DA82AB3D8D80AE0F4F05A; Path=/
Location: http://www.willselfdestruct.com/secure/success
Content-Length: 0
Connection: close
Content-Type: text/plain; charset=UTF-8

```

POST requests are how form data gets submitted to a web server, GET requests only ask for a page, POST requests send data.

When filtering for every POST in the capture, it gives a shortlist of candidate submissions; cross-reference the destination host/IP against the willselfdestruct.com IP to identify the specific POST that represents the harassing message actually being sent.

Item	My Findings
Packet number of the relevant POST	83601
TCP stream number	1707
Destination host/IP (willselfdestruct.com)	www.willselfdestruct.com (69.25.94.22)
Form field(s) identified	to, from (blank), subject, message, type, ttl, submit.x/submit.y (button click coordinates)
How the content correlates with the reported message	The to field is literally lilytuckrige@yahoo.com, an exact match to the reported victim. The subject ("you can't find us") and message body ("and you can't hide from us... Stop teaching... Start running.") are directly threatening/harassing in nature, consistent with Lily Tuckrige's complaint of receiving harassing emails. The server's 302 redirect to /secure/success confirms the submission completed successfully from this client at this exact timestamp.

8. Part D - Identify the Device (MAC Address)

18815	2008-07-22T00:29:51.677648000-0400	00:17:f2:e2:c0:ce	00:1d:d9:2e:4f:60	192.168.15.4	192.168.1.254
18816	2008-07-22T00:29:51.678746000-0400	00:1d:d9:2e:4f:60	00:17:f2:e2:c0:ce	192.168.1.254	192.168.15.4
18818	2008-07-22T00:29:53.706234000-0400	00:17:f2:e2:c0:ce	00:1d:d9:2e:4f:60	192.168.15.4	192.168.1.254
18819	2008-07-22T00:29:53.719600000-0400	00:1d:d9:2e:4f:60	00:17:f2:e2:c0:ce	192.168.1.254	192.168.15.4
18820	2008-07-22T00:29:53.728711000-0400	00:17:f2:e2:c0:ce	00:1d:d9:2e:4f:60	192.168.15.4	72.21.210.11
18821	2008-07-22T00:29:53.809869000-0400	00:1d:d9:2e:4f:60	00:17:f2:e2:c0:ce	72.21.210.11	192.168.15.4
18822	2008-07-22T00:29:53.812119000-0400	00:17:f2:e2:c0:ce	00:1d:d9:2e:4f:60	192.168.15.4	72.21.210.11
18823	2008-07-22T00:29:53.812853000-0400	00:17:f2:e2:c0:ce	00:1d:d9:2e:4f:60	192.168.15.4	72.21.210.11
18824	2008-07-22T00:29:53.902967000-0400	00:1d:d9:2e:4f:60	00:17:f2:e2:c0:ce	72.21.210.11	192.168.15.4
18826	2008-07-22T00:29:54.673425000-0400	00:1d:d9:2e:4f:60	00:17:f2:e2:c0:ce	72.21.210.11	192.168.15.4
18827	2008-07-22T00:29:54.675352000-0400	00:1d:d9:2e:4f:60	00:17:f2:e2:c0:ce	72.21.210.11	192.168.15.4
18828	2008-07-22T00:29:54.677607000-0400	00:1d:d9:2e:4f:60	00:17:f2:e2:c0:ce	72.21.210.11	192.168.15.4
18829	2008-07-22T00:29:54.678062000-0400	00:17:f2:e2:c0:ce	00:1d:d9:2e:4f:60	192.168.15.4	72.21.210.11
18830	2008-07-22T00:29:54.720260000-0400	00:17:f2:e2:c0:ce	00:1d:d9:2e:4f:60	192.168.15.4	72.21.210.11
18831	2008-07-22T00:29:54.761277000-0400	00:1d:d9:2e:4f:60	00:17:f2:e2:c0:ce	72.21.210.11	192.168.15.4
18832	2008-07-22T00:29:54.763342000-0400	00:1d:d9:2e:4f:60	00:17:f2:e2:c0:ce	72.21.210.11	192.168.15.4
18833	2008-07-22T00:29:54.765294000-0400	00:17:f2:e2:c0:ce	00:1d:d9:2e:4f:60	192.168.15.4	72.21.210.11
18834	2008-07-22T00:29:54.765499000-0400	00:1d:d9:2e:4f:60	00:17:f2:e2:c0:ce	72.21.210.11	192.168.15.4
18835	2008-07-22T00:29:54.793346000-0400	00:17:f2:e2:c0:ce	00:1d:d9:2e:4f:60	192.168.15.4	192.168.1.254
18836	2008-07-22T00:29:54.803603000-0400	00:1d:d9:2e:4f:60	00:17:f2:e2:c0:ce	72.21.210.11	192.168.15.4
18837	2008-07-22T00:29:54.805672000-0400	00:1d:d9:2e:4f:60	00:17:f2:e2:c0:ce	72.21.210.11	192.168.15.4
18838	2008-07-22T00:29:54.806130000-0400	00:17:f2:e2:c0:ce	00:1d:d9:2e:4f:60	192.168.15.4	72.21.210.11
18839	2008-07-22T00:29:54.814050000-0400	00:1d:d9:2e:4f:60	00:17:f2:e2:c0:ce	192.168.1.254	192.168.15.4
18840	2008-07-22T00:29:54.827712000-0400	00:17:f2:e2:c0:ce	00:1d:d9:2e:4f:60	192.168.15.4	69.22.167.225
18841	2008-07-22T00:29:54.828131000-0400	00:17:f2:e2:c0:ce	00:1d:d9:2e:4f:60	192.168.15.4	69.22.167.225
18842	2008-07-22T00:29:54.836991000-0400	00:1d:d9:2e:4f:60	00:17:f2:e2:c0:ce	69.22.167.225	192.168.15.4
18843	2008-07-22T00:29:54.838229000-0400	00:1d:d9:2e:4f:60	00:17:f2:e2:c0:ce	69.22.167.225	192.168.15.4
18844	2008-07-22T00:29:54.839565000-0400	00:17:f2:e2:c0:ce	00:1d:d9:2e:4f:60	192.168.15.4	69.22.167.225
18845	2008-07-22T00:29:54.839922000-0400	00:17:f2:e2:c0:ce	00:1d:d9:2e:4f:60	192.168.15.4	69.22.167.225
18846	2008-07-22T00:29:54.840726000-0400	00:17:f2:e2:c0:ce	00:1d:d9:2e:4f:60	192.168.15.4	69.22.167.225

```
tshark -r working/nitroba_working.pcap -Y "ip.addr == 192.168.15.4" -T fields \
-e frame.number -e frame.time -e eth.src -e eth.dst -e ip.src -e ip.dst \
| head -n 30 | tee reports/client_mac_sample.tsv
```

This, samples traffic to or from the client IP to identify its consistent Ethernet (MAC) address, the actual network interface card that sent the traffic, one layer below the IP address.

94374	2008-07-22T02:13:05.469430000-0400	192.168.15.4	12.129.210.46	TCP	35944 → 80 [RST, ACK] Seq=2937 Ack=4798 Win=65535 Len=0
94385	2008-07-22T02:13:14.997080000-0400	192.168.15.4	74.125.19.99	TCP	[TCP Retransmission] 36858 → 80 [FIN, ACK] Seq=514 Ack=594 Win=524280 Len=0 TSval=936588612 TSecr=1001670443
94386	2008-07-22T02:13:29.918742000-0400	192.168.15.4	64.128.80.61	TCP	[TCP Retransmission] 36414 → 80 [FIN, ACK] Seq=391 Ack=1576 Win=524280 Len=0 TSval=936588761 TSecr=3948817815
94387	2008-07-22T02:13:35.607017000-0400	209.85.171.190	192.168.15.4	TCP	443 → 36862 [FIN, ACK] Seq=1584 Ack=1303 Win=9728 Len=0 TSval=998711333 TSecr=936587620
94388	2008-07-22T02:13:35.677031000-0400	192.168.15.4	209.85.171.190	TCP	36862 → 443 [ACK] Seq=1303 Ack=1585 Win=524280 Len=0 TSval=936588819 TSecr=998711333
94389	2008-07-22T02:13:35.837079000-0400	38.102.35.112	192.168.15.4	TCP	80 → 36458 [RST, ACK] Seq=493 Ack=2335 Win=0 Len=0
94390	2008-07-22T02:13:39.234514000-0400	192.168.15.4	84.200.227.131	TCP	[TCP Retransmission] 36822 → 80 [FIN, ACK] Seq=836 Ack=47910 Win=524280 Len=0 TSval=936588854 TSecr=211816
94391	2008-07-22T02:13:39.234657000-0400	192.168.15.4	84.200.227.131	TCP	[TCP Retransmission] 36820 → 80 [FIN, ACK] Seq=115 Ack=1058 Win=524280 Len=0 TSval=936588854 TSecr=211827
94392	2008-07-22T02:13:41.737325000-0400	192.168.15.4	84.200.227.131	TCP	[TCP Retransmission] 36172 → 80 [FIN, ACK] Seq=758 Ack=3343 Win=524280 Len=0 TSval=936588879 TSecr=179695
94393	2008-07-22T02:13:41.737581000-0400	192.168.15.4	84.200.227.131	TCP	[TCP Retransmission] 36166 → 80 [FIN, ACK] Seq=773 Ack=921 Win=524280 Len=0 TSval=936588879 TSecr=179693
94394	2008-07-22T02:13:41.737859000-0400	192.168.15.4	84.200.227.131	TCP	[TCP Retransmission] 36164 → 80 [FIN, ACK] Seq=152 Ack=73292 Win=524280 Len=0 TSval=936588879 TSecr=179764
94396	2008-07-22T02:13:44.266370000-0400	192.168.15.4	209.85.171.190	TLSv1	Encrypted Alert
94397	2008-07-22T02:13:44.266638000-0400	192.168.15.4	209.85.171.190	TCP	36862 → 443 [FIN, ACK] Seq=1330 Ack=1585 Win=524280 Len=0 TSval=936588905 TSecr=998711333
94398	2008-07-22T02:13:44.293692000-0400	209.85.171.190	192.168.15.4	TCP	443 → 36862 [RST] Seq=1585 Win=0 Len=0
94410	2008-07-22T02:13:47.046029000-0400	192.168.15.4	74.125.19.99	TCP	[TCP Retransmission] 36858 → 80 [FIN, ACK] Seq=514 Ack=594 Win=524280 Len=0 TSval=936588932 TSecr=1001670443

```
tshark -r working/nitroba_working.pcap -Y "eth.addr == 00:17:f2:e2:c0:ce" -T fields \
-e frame.number -e frame.time -e ip.src -e ip.dst -e _ws.col.Protocol -e _ws.col.Info \
| tee reports/client_device_activity.tsv
```

This, traces every packet from that specific device across the whole capture, regardless of momentary IP changes (relevant on DHCP networks). This matters because IP addresses can be reassigned or shared (exactly the concern flagged in this case's shared open-wireless setup), while a MAC address ties activity to one physical network interface, though even a MAC address only identifies a device, not who was sitting at it.

Item	Findings
	submission
Caveats on this evidence	A cookie reflects what a browser session was authenticated as at that moment, it shows the device had an active Gmail login, not conclusively that the same person who was logged into Gmail was also the one who submitted the harassment form minutes/hours earlier or later. It also doesn't rule out multiple people using the same shared device/browser profile on this open network.

10. Part F - Cross-Reference the Class Roster

1. Take the identity suggested by Part E's cookie/browser evidence.
2. Compare it against the supplied Chemistry 109 class roster from the scenario materials.
3. Document whether there is a match, and the exact basis for that comparison (e.g., matching name, matching email format/domain).
4. Note explicitly if the roster comparison is itself limited — e.g., if multiple students could plausibly share initials, or if the roster doesn't include a directly comparable identifier.

```

└─$ wc -l working/slides_text.txt
grep -n -i "roster\\|chem109\\|class list\\|student" working/slides_text.txt
73 working/slides_text.txt
8:Lily Tuckrige is teaching chemistry CHEM109 this summer at NSU.
11:She thinks that it is from one of the students in her class.
25:34.62.247.140.in-addr.arpa domain name pointer G24.student.nitroba.org
49:Chemistry 109 class list:
53:Students:

```

```

└─(kali㉿kali)-[~/SBT-DF204-Case1]
└─$ sed -n '45,73p' working/slides_text.txt

```

And here is the message:

No, here is the message

And there goes the message:

So who did it?

Chemistry 109 class list:

Teacher: Lily Tuckrige

Students:

```

    Amy Smith
    Burt Greedom
    Tuck Gorge
    Ava Book
    Johnny Coach
    Jeremy Ledvkin

```



```
(kali@kali)-[~/SBT-DF204-Case1]
$ sed -n '45,73p' working/slides_text.txt
```

And here is the message:

No, here is the message

And there goes the message:

So who did it?

Chemistry 109 class list:

Teacher: Lily Tuckrige

Students:

- Amy Smith
- Burt Greedom
- Tuck Gorge
- Ava Book
- Johnny Coach
- Jeremy Ledvkin
- Nancy Colburne
- Tamara Perkins
- Esther Pringle
- Asar Misrad
- Jenny Kant

How to solve this problem:

Map out the Nitroba dorm room network.

Find who sent email to lilytuckrige@yahoo.com

Look for a TCP flow that includes the hostile message

Find information that can tie that message to a particular web browser.

Identify the other TCP connections that belong to the attacker

Find information in one of those TCP connections that IDs the attacker.

Title 8 Bullets (nitroba)

Item	Findings
Identity from Part E	jcoachj@gmail.com
Roster match found?	Yes
Basis for the match	The Gmail username "jcoachj" directly corresponds to student "Johnny Coach" on the Chemistry 109 roster, first-initial ("j") + surname ("coach") is a standard email-naming convention, and no other roster entry produces a comparable match
Limitations of this comparison	The match is based on a plausible naming convention inference, not a confirmed registration record tying the Gmail account to Johnny Coach by name. The roster itself does not list student email addresses directly, so this remains circumstantial, though it is a

Item	Findings
	strong, specific match with no competing candidates among the eleven students listed

10. Part G - Build the Timeline

Time (as shown in Wireshark)	Packet #	Event	Source
2008-07-22T02:03:43.825871000-0400	82936	Client loads willselfdestruct.com submission page (GET /secure/submit)	Part A
2008-07-22T02:03:44.210946000-0400	82998	Page content confirms "secure anonymous E-mail" service branding	Part B
2008-07-22T02:04:24.311700000-0400	83601	HTTP POST submits harassment message to willselfdestruct.com	Part C
2008-07-22T02:01:13.690666000-0400	19652	Gmail session cookie observed (jcoachj@gmail.com) on client device	Part E
2008-07-22T02:04:24.564165000-0400	83614	Server confirms success (302 redirect to /secure/success)	Part C

12. Part H - Attribution Assessment

This is the highest-weighted section in the marking guide alongside message correlation. And i structured it's explicitly into three categories:

Category	Content
Direct observations (what the packets literally show)	Client device 192.168.15.4 (MAC 00:17:f2:e2:c0:ce) loaded the willselfdestruct.com submission page at 02:03:43 (frame 82936) and submitted an HTTP POST at 02:04:24 (frame 83601, TCP stream 1707) containing the message "to=lilytuckrige@yahoo.com... subject=you can't find us... message=and you can't hide from us... Stop teaching... Start running." The server returned a 302 success response (frame 83614), and both the page and success response contained the service's "secure anonymous E-mail" branding text (frames 82998,

Category	Content
	83641). The same device carried Gmail session cookies at 02:01:13-02:01:17 (frames 19652-19732) containing the identifier jcoachj@gmail.com. The Chemistry 109 class roster lists a student named Johnny Coach, whose name maps directly to the "jcoachj" username under a standard first-initial-plus-surname email convention; no other roster entry produces a comparable match.
Reasonable inferences (what you conclude, and why)	The close timing (under 3.5 minutes) between the Gmail-session cookie evidence and the harassment POST, both from the same device, combined with the specific and uncontested match between "jcoachj" and roster entry "Johnny Coach," supports a reasonable inference that the device used to submit the harassment message was, around that time, associated with a Gmail account belonging to or used by a student matching this roster entry. This is an inference built from converging circumstantial evidence, not a direct, confirmed identification.
Limitations / alternative explanations (at least two)	1) Shared open wireless network means the IP/MAC-level evidence identifies a device on the network, not conclusively a specific person at the keyboard at that moment, the router (00:1d:d9:2e:4f:60) serves a residence with multiple occupants. 2) Cookie evidence reflects authenticated browser-session state, not physical presence, a Gmail session can remain logged in on a shared or unattended device, so the cookie shows who the browser was authenticated as minutes earlier, not conclusively who typed the POST. 3) The roster match is a naming-convention inference, not a confirmed account-ownership record, the roster does not list student email addresses directly, so "jcoachj" has not been verified against an official university account record for Johnny Coach.

Confidence level: Moderate. The evidence chain is specific, internally consistent, and uncontested by any competing candidate on the roster, but it rests on inference (username-pattern matching and session timing) rather than a confirmed, independently verified identity record, which is why this falls short of "high" confidence.

13. Evidence Log

Evidence ID	Packet/Item	Finding	Why It Matters
E01	Original & working file hashes	nitroba.pcap, SHA-256: 2b77a9eaefc1d6af163d1ba793c96dbccacb04e6befdf1a0b01f8c67553ec2fb (original = working copy)	Establishes evidence integrity and chain of custody
E02	Frame 82936	Client 192.168.15.4 GET /secure/submit to www.willselfdestruct.com at 02:03:43	First confirmed contact with the self-destructing message service
E03	Frame 83601, TCP stream 1707	HTTP POST containing to=lilytuckrige@yahoo.com and threatening message text	Direct evidentiary link between this client and the harassment message content
E04	Frame 83614	302 redirect to /secure/success	Confirms the harassment message submission completed successfully
E05	eth.addr 00:17:f2:e2:c0:ce (73,246 matching frames)	Client device MAC consistently associated with IP 192.168.15.4 throughout capture	Ties the harassment submission to one specific physical network device
E06	Frames 19652-19732	Gmail session cookies showing gmailchat=jcoachj@gmail.com on the same device, ~3 min before the POST	

14. Executive Summary

This case study report examines network traffic captured at a university residence boundary (nitroba.pcap, SHA-256:

2b77a9eaefc1d6af163d1ba793c96dbccacb04e6befdf1a0b01f8c67553ec2fb, confirmed matching the official checksum published by digitalcorpora.org) in relation to a reported harassment complaint. Analysis confirms that a client at IP 192.168.15.4 (MAC 00:17:f2:e2:c0:ce) accessed the self-destructing-message service www.willselfdestruct.com at 02:03:43 (packet #82936) and submitted an HTTP POST at 02:04:24 (packet #83601) whose content, addressed to **lilytuckrige@yahoo.com**, with a threatening subject line and message body, is consistent with the reported harassing message.

Gmail session cookie evidence observed on the same device approximately three minutes earlier (packets #19652-19732, 02:01:13-02:01:17) suggests an association with the account **jcoachj@gmail.com**, which corresponds to the Chemistry 109 roster entry "Johnny Coach" under a standard first-initial plus surname email naming convention, with no competing match among the eleven listed students.

This evidence supports a moderate confidence attribution to the student Johnny Coach, based on the Gmail account **jcoachj@gmail.com** observed active on the submitting device shortly before the harassment message was sent. This conclusion is limited by the fact that the originating network is a shared, open wireless router serving multiple residents, meaning the IP address alone cannot establish which individual was using a given device at the time of the request. Additional limitations include: the Gmail cookie evidence reflects an authenticated browser session at a point in time rather than confirmed physical presence at the keyboard three minutes later when the message was actually submitted, and the roster match itself is a naming convention inference rather than a verified account record.

No further identification beyond what is stated above is supported by this capture alone.